

Research Statement

Project: *Trustworthy AI Systems: Enhancing Robustness and Safety in Adversarial Environments*

Reference: [TODO: verify with user]

Host: ELLIS Institute Tübingen / Max Planck Institute for Intelligent Systems

Supervisor: Dr. Maksym Andriushchenko

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Abstract

The rapid advancement of artificial intelligence (AI) systems has brought about significant benefits across various domains. However, the increasing complexity and autonomy of these systems have raised concerns regarding their safety and robustness, particularly in adversarial settings. This research aims to address these concerns by developing a framework for trustworthy AI systems that can withstand adversarial attacks and maintain alignment with human values. Building on the foundational work of Dr. Maksym Andriushchenko in AI safety and adversarial robustness, this study will explore the integration of adversarial training techniques, robust optimisation, and safety evaluations to enhance the resilience of AI models. The research will also investigate the use of large language models (LLMs) for generating adaptive decision rules that improve system robustness in dynamic environments. By focusing on both technical and behavioural aspects, this project seeks to contribute to the development of AI systems that are not only technically sound but also socially responsible and aligned with human values.

1. Introduction and Problem Statement

The deployment of AI systems in critical applications necessitates a focus on their safety and robustness, especially in adversarial environments where malicious actors may attempt to exploit system vulnerabilities. Dr. Maksym Andriushchenko's recent work highlights the importance of adversarial robustness and trustworthy machine learning (Andriushchenko et al., 2023; Andriushchenko et al., 2024; Andriushchenko et al., 2025). Despite advancements in adversarial training and robust optimisation, challenges remain in ensuring that AI systems can effectively resist adversarial attacks while maintaining alignment with human values. This research addresses these challenges by proposing a comprehensive framework that integrates adversarial robustness techniques with safety evaluations and behavioural modelling.

2. Research Questions

Three interconnected research questions guide this study:

RQ1, Adversarial Robustness. How can adversarial training techniques be optimised to enhance the robustness of AI systems against a wide range of adversarial attacks?

RQ2, Safety Evaluations. What methodologies can be employed to systematically evaluate the safety and alignment of AI systems with human values in dynamic and adversarial environments?

RQ3, Behavioural Modelling. How can large language models be utilised to generate adaptive decision rules that improve the robustness and safety of AI systems in real-world scenarios?

3. Literature Review

Adversarial Robustness. The field of adversarial robustness has seen significant progress, with techniques such as adversarial training and robust optimisation being widely studied (Goodfellow et al., 2015; Madry et al., 2018). Dr. Andriushchenko's work has contributed to this area by exploring novel adversarial training methods that enhance model resilience (Andriushchenko et al., 2023; Andriushchenko et al., 2024). However, challenges remain in developing models that can withstand diverse and evolving adversarial strategies.

Safety Evaluations. Ensuring the safety and alignment of AI systems with human values is a critical concern. Recent studies have proposed various safety evaluation frameworks, but there is a need for more comprehensive approaches that consider both technical

and behavioural aspects (Amodei et al., 2016; Leike et al., 2018). Dr. Andriushchenko's research in trustworthy machine learning provides a foundation for developing such frameworks (Andriushchenko et al., 2025).

Behavioural Modelling. The use of large language models for generating adaptive decision rules is an emerging area of research. Studies have shown the potential of LLMs in improving system robustness by adapting to dynamic environments (Brown et al., 2020; Radford et al., 2021). This research will build on these findings to explore the integration of LLMs with adversarial robustness techniques.

4. Methodology

4.1 Overview

The research will follow a three-stage pipeline to develop and evaluate the proposed framework:

flowchart LR

A[Stage 1: Adversarial Training] --> B[Stage 2: Safety Evaluations] --> C[Stage 3: Behavioural Modelling]

4.2 Quantitative

The quantitative component will involve the development and testing of adversarial training techniques. This will include the implementation of robust optimisation algorithms and the evaluation of model performance against various adversarial attacks.

4.3 Qualitative

Qualitative methods will be employed to assess the alignment of AI systems with human values. This will involve stakeholder interviews and surveys to gather insights into the ethical and social implications of AI deployment.

4.4 Interplay

The interplay between quantitative and qualitative methods will be explored through the integration of large language models. These models will be used to generate adaptive decision rules that enhance system robustness while maintaining alignment with human values.

5. Expected Outcomes and Significance

The research is expected to yield a comprehensive framework for developing trustworthy AI systems that are robust against adversarial attacks and aligned with human values. By integrating adversarial training techniques with safety evaluations and behavioural modelling, the study aims to contribute to the advancement of AI safety and alignment research. The findings will have implications for the design and deployment of AI systems in critical applications, ensuring their reliability and trustworthiness.

6. Fit with Existing Background

My MSc dissertation on enhancing supply chain information sharing systems through blockchain and LLM analysis (Zulfikar, 2024) aligns with the proposed research by demonstrating my ability to integrate technical and behavioural aspects in complex systems. Additionally, my GitHub projects, such as the LLM-Generated Adaptive Shipper Decision Rules and the Mini Digital Twin, showcase my expertise in developing robust and adaptive AI models (nauvalZulfikar, 2026).

7. Three-Year Workplan

- **Y1:**
 - Conduct literature review and identify key challenges in adversarial robustness and AI safety.
 - Develop initial adversarial training techniques and evaluate model performance.
 - Begin qualitative assessments of AI system alignment with human values.

- **Y2:**
 - Refine adversarial training methods and integrate safety evaluation frameworks.
 - Conduct stakeholder interviews and surveys to gather insights into ethical implications.
 - Develop and test large language models for adaptive decision rule generation.
 - **Y3:**
 - Finalise the comprehensive framework for trustworthy AI systems.
 - Conduct extensive testing and validation of the proposed framework.
 - Disseminate findings through publications and presentations at conferences.
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8. Challenges and Limitations

1. **Complexity of Adversarial Attacks:** The evolving nature of adversarial attacks may pose challenges in developing robust models. Mitigation: Regularly update training datasets and algorithms to address new attack vectors.
 2. **Ethical Considerations:** Balancing technical advancements with ethical implications can be challenging. Mitigation: Engage with diverse stakeholders to ensure ethical considerations are integrated into the research.
 3. **Resource Constraints:** Limited computational resources may hinder large-scale experiments. Mitigation: Leverage cloud computing platforms and collaborate with research institutions for access to necessary resources.
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9. Conclusion and Why ELLIS Institute Tübingen

The ELLIS Institute Tübingen, with its focus on AI safety and alignment, provides an ideal environment for conducting this research. Dr. Maksym Andriushchenko's expertise in adversarial robustness and trustworthy machine learning aligns perfectly with the proposed study. The institute's commitment to interdisciplinary research and collaboration will support the development of innovative solutions to enhance the safety and robustness of AI systems.

References

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